MSc in Mathematics and Foundations of Computer Science

CATEGORIES, PROOFS AND PROCESSES

Michaelmas Term 2025

Submission deadline 12 noon, Wednesday 7th January 2026, via Inspera.

There is a total of 100 marks available for this paper, you should attempt all parts of the paper.

NB: You must not discuss this examination paper with anyone.

TURN OVER

This paper consists of two questions. You should attempt both of them (note that Q2 is worth considerably more marks than Q1). Provide justification for all your answers. Your answers will be judged by their correctness and completeness as well as by the clarity of presentation.

You are free to rely on any results known from the lectures and/or the lecture notes, without proof. If you rely on any other sources, you must explicitly declare and reference any such use.

Question 1

In this question we investigate various notions of monicity for natural transformations.

Recall that in any category $\mathcal{C}$, an arrow $f : A \to B$ is called a *split mono* if there exists an arrow $g : B \to A$ such that $g \circ f$ is the identity arrow on A. Also, call an arrow a *regular mono* if it is an equaliser of some pair of parallel arrows in $\mathcal{C}$. We know from the lectures and problem sheets that there are some implications between monicity properties that hold in every category:

$$\text{split mono} \implies \text{regular mono} \implies \text{mono}$$

For any property P of arrows, a natural transformation $\alpha : F \to G$ (where $F, G : \mathcal{C} \to \mathcal{D}$) can be called:

- *naturally* P, if it has the property P as an arrow in the functor category $\mathcal{D}^{\mathcal{C}}$,
- *pointwise* P, if for every object X in $\mathcal{C}$, $\alpha_X : FX \to GX$ has the property P as an arrow in $\mathcal{D}$.

For each $P \in \{\text{split mono}, \text{regular mono}, \text{mono}\}$, determine which implications hold in general between the properties *naturally* P and *pointwise* P of natural transformations. (To clarify: there are six implications to consider, each worth 4 marks. Prove or find a counterexample for each of them.)

(24 marks)

Question 2

In this question we investigate a few categories of algebras. We begin with some definitions.

A *signature* is a (possibly empty) set Σ, whose elements we call function symbols, together with a function $ar : \Sigma \to \mathbb{N}$, which assigns a finite (possibly zero) arity to each symbol. A Σ-*algebra*

$$\mathcal{A} = \left(A, \left(f^{\mathcal{A}}\right)_{f \in \Sigma}\right)$$

is a set A (the *domain* of $\mathcal{A}$) together with, for every function symbol $f \in \Sigma$, a function

$$f^{\mathcal{A}} : A^{ar(f)} \to A.$$

For Σ-algebras $\mathcal{A}$ and $\mathcal{B}$ (with domains respectively A and B), a *homomorphism* from $\mathcal{A}$ to $\mathcal{B}$ is a function $h : A \to B$ such that

$$h(f^{\mathcal{A}}(a_1, \ldots, a_n)) = f^{\mathcal{B}}(h(a_1), \ldots, h(a_n))$$

for every $f \in \Sigma$ and $a_1, \ldots, a_n \in A$, where $n = ar(f)$.

For an algebra $\mathcal{A}$, an element $k \in A$ is a *kixpoint* if, for every $f \in \Sigma$, either $f^{\mathcal{A}}(k, k, \ldots, k) \neq k$ or $f^{\mathcal{A}}$ is a constant function that always returns k (i.e. $f^{\mathcal{A}}(a_1, \ldots, a_n) = k$ for all $a_i \in A$).

For any signature Σ, consider the following three categories:

- $\mathbf{Alg}_\Sigma$, the category of all Σ-algebras and their homomorphisms,
- $\mathbf{KAlg}_\Sigma$, the full subcategory of $\mathbf{Alg}_\Sigma$ that contains all Σ-algebras with at least one kixpoint,
- $\mathbf{XAlg}_\Sigma$, the category where objects are pairs $(\mathcal{A}, k)$ where $\mathcal{A}$ is a Σ-algebra and k is a kixpoint in $\mathcal{A}$, and arrows $h : (\mathcal{A}, k) \to (\mathcal{B}, l)$ are homomorphisms $h : \mathcal{A} \to \mathcal{B}$ such that $h(k) = l$.

For each of the three categories above determine whether, for every signature Σ, it has:

(a) initial objects, (9 marks)

(b) terminal objects, (6 marks)

(c) (binary) products, (9 marks)

(d) (binary) coproducts, (12 marks)

(e) equalisers, (9 marks)

(f) coequalisers. (15 marks)

Moreover:

(g) There are evident forgetful functors: (i) from $\mathbf{XAlg}_\Sigma$ to $\mathbf{KAlg}_\Sigma$, $\mathbf{Alg}_\Sigma$ and $\mathbf{Set}$; (ii) from $\mathbf{KAlg}_\Sigma$ to $\mathbf{Set}$, and (iii) from $\mathbf{Alg}_\Sigma$ to $\mathbf{Set}$. For each of these, and for the inclusion functor from $\mathbf{KAlg}_\Sigma$ to $\mathbf{Alg}_\Sigma$, determine whether it has a left adjoint for every Σ. (16 marks)

Positive answers to all parts should be accompanied with explicit constructions.

In all parts of this question, you may aim for partial credit (up to 50% of the maximum mark) by considering the special case where $ar(f) = 1$ for every f in Σ. Please make this decision explicit for every part where you choose to do so.

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